Schedule

Practice problems/reading assignments on HT: on LearnHub

Lab 5: on LearnHub

Case Exam 1: Saturday, April 19th at 18:30-21:00 in labs SNAB149 and SNAB242

- Seating charts will be announced
- Only lab computers can be used
- Safe Exam Browser; Phyton, Excel, common report area

Will have access to LearnHub material

Midterm: Friday, April 25th at 19:00-21:00, on LearnHub, multiple choice, in labs SNAB149 and SNAB220

- Safe Exam Browser, only Excel can be used
- Seating charts will be announced

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Agenda

Labs

Hypothesis testing: short review

Class exercise

Significance testing

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Lab 2

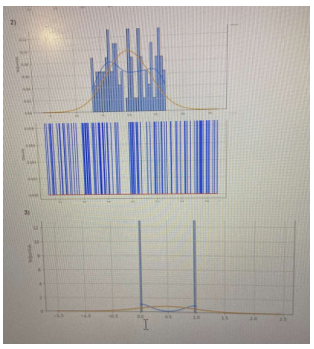
1) Compute the population mean, median, and standard deviation of random variables in part A. Compare these with the corresponding statistics of the samples in part A.
2) Draw the PDF/PMF of random variables in part A. Compare these functions' shapes with the plots you generate in part A.

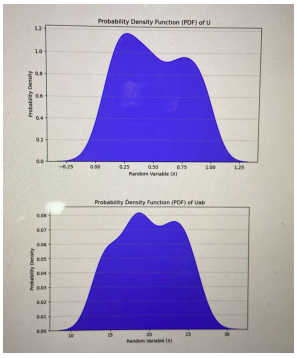
Random variables in Part A:
- uniform random variables between [0,1]
- uniform random variables between [a=13, b=27]
- fair coin tosses (0 is tails and 1 is heads, or vice versa)
- unfair coin tosses (probability of heads = 0.7, probability of tails = 0.3)
- normal random variable with mean μ=50 and σ=10

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Lab 2

Draw the PDF/PMF of random variables in part A. Compare these functions' shapes with the plots you generate in part A.

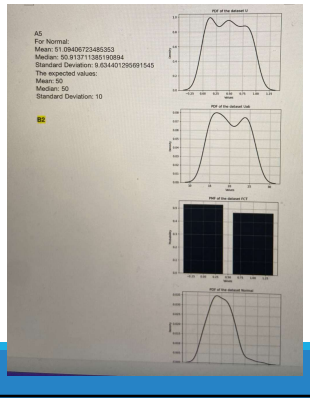


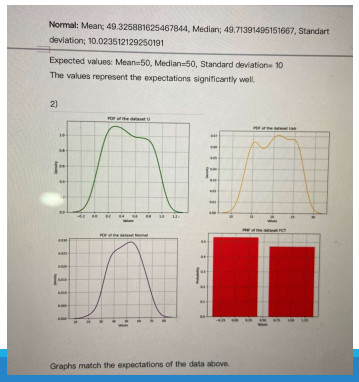


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Lab 2

Draw the PDF/PMF of random variables in part A. Compare these functions' shapes with the plots you generate in part A.





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Lab assignments

- The idea was to give good grades to effort with grades of 0, 7, 10.
- However, in the presence of intense AI use, it isn't easy to measure effort.
- Now the assignments are TurnItIn assignments mainly to check AI use.
- The grades are spread to (0,10).
- No change in the previous grades.

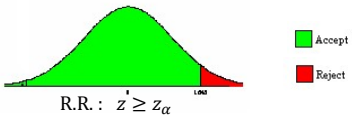
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Testing population mean of a normal r.v. with large sample

H_0: $\mu = \mu_0$
H_a: $\mu > \mu_0$

Reject H_0 if $\bar{X}$ is large enough. $\longrightarrow$ $Z = \frac{\bar{X} - \mu}{S/\sqrt{n}} \geq z_\alpha \Leftrightarrow \bar{X} \geq \mu + z_\alpha \frac{S}{\sqrt{n}}$

Test statistics: $Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$ $\longrightarrow$



R.R. : $z \geq z_\alpha$

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Example: Milemaster tires

Type I is serious, so we can set a bound on probability of type-I error.
Using this bound we can find the value of "large enough" to reject H_0.

α : significance level

$\alpha = \mathrm{P}\{\text{Type I}\} = \mathrm{P}\{\text{reject } \mathrm{H_0} \text{ when } \mathrm{H_0} \text{ is true}\}$

$= P\{\bar{X} > k, \mu = 47000\} = P\{\frac{\bar{X} - \mu}{S/\sqrt{n}} > \frac{k - \mu_0}{S/\sqrt{n}}\}$

$= P\{Z > \frac{k - \mu_0}{S/\sqrt{n}}\} = P\{Z > \frac{k - 47000}{1924.56/\sqrt{50}}\} = P\{Z > z_\alpha\}$

Let α = 0.05. Do we reject the null hypothesis for $\bar{x} = 47{,}351$ miles?

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Relation between α and k

Type I is serious, so we can set a bound on probability of type-I error.
Using this bound we can find the value of "large enough" to reject H_0.

α : significance level

$\alpha = \mathrm{P}\{\text{Type I}\} = \mathrm{P}\{\text{reject } \mathrm{H_0} \text{ when } \mathrm{H_0} \text{ is true}\}$

$$\frac{k-\mu_0}{S/\sqrt{n}} = z_\alpha \Rightarrow k = \mu_0 + z_\alpha \frac{S}{\sqrt{n}}$$

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Example: Milemaster tires

We know the probability of type-I error. How about probability of type-II error?

$$\beta = \mathrm{P}\{\text{Type II}\} = \mathrm{P}\{\text{cannot reject } \mathrm{H_0} \text{ when } \mathrm{H_0} \text{ is false}\}$$

β can be calculated only at a given value of μ s.t. $\mu > 47{,}000$.

$$\beta = \mathrm{P}\{\bar{\mathrm{X}} < \mathrm{k}, \mu = 47500\} = \mathrm{P}\{\frac{\bar{X} - \mu_1}{S/\sqrt{n}} < \frac{k - \mu_1}{S/\sqrt{n}}\} = P\{Z < \frac{k - \mu_1}{S/\sqrt{n}}\} = \ldots$$

$$\beta = \mathrm{P}\{\bar{\mathrm{X}} < \mathrm{k}, \mu = 57000\} = \ldots$$

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Tests about a population mean

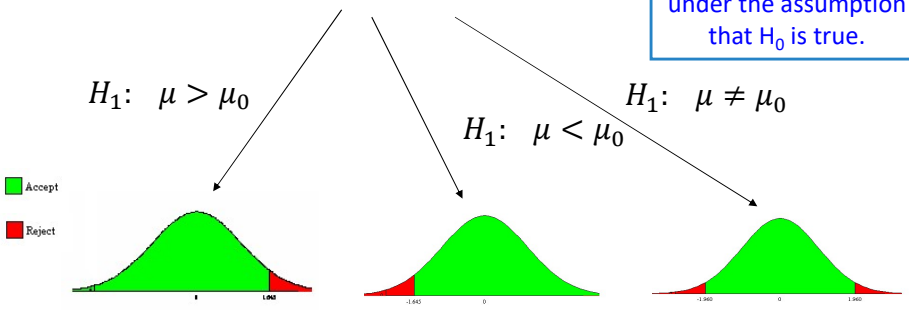
Upper tailed test	***Lower tailed test***	***Two- tailed test***
H_0: $\mu = \mu_0$	H_0: $\mu = \mu_0$	H_0: $\mu = \mu_0$
H_a: $\mu > \mu_0$	H_a: $\mu < \mu_0$	H_a: $\mu \neq \mu_0$
An estimator for μ = ?	$\bar{X}$	**How to use $\bar{X}$ to conclude about these tests?**
$\bar{X} > k_U$	$\bar{X} < k_L$	$\bar{X} < k'_L$ $\bar{X} > k'_U$

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Testing population mean of a normal r.v. with large sample

H_0: $\mu = \mu_0$ Test statistics: $Z = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$

Test statistics is given under the assumption that H_0 is true.



R.R. : $z \geq z_\alpha$ R.R. : $z \leq -z_\alpha$ R.R. : $z \leq -z_{\alpha/2}$ or $z \geq z_{\alpha/2}$

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Analysis of hypotheses testing-I

The truth and falsity of a particular process can never be known with certainty!! (unless you examine the whole population)

Therefore: a hypotheses-testing procedure should be developed with the probability of reaching a wrong conclusion in mind!

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Analysis of hypotheses testing - II

H_0 will be rejected in favor of H_a only if sample evidence suggests H_0 is false.

If the sample does not strongly contradict H_0, we continue to believe in the truth of H_0.

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When to formulate these hypotheses?

EXTERNAL EFFECTS (to the statistical process)

- A new product will be introduced: Cost of the product
- Other examples: Certain specifications needed, etc.

CHANGE IN THE STATISTICAL PROCESS

- More complaints from your long-term customers on your product: HAS THE DEFECT PROPORTION CHANGE?
- Are the fish safe? (HAS STH. CHANGED IN THE FISH?)

RESULT OF PHYSICAL/CHEMICAL MODELS

- the effect of increasing the amount of iron in a certain alloy on the tensile strength.

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Computation of errors

α = P{Type I} = P{reject H_0 when H_0 is true}

α : called significance level

β = P{Type II} = P{accept H_0 when H_0 is false}

1- β : power of the test

1- β = P{reject H_0 when H_0 is false}

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Effects of errors

Type-I error is more serious than type-II error.

Hypotheses are chosen accordingly.

Approach:

- Specify the largest value of α, that can be tolerated.
- Find a rejection region satisfying that α.
- This procedure makes β as small as possible subject to a bound on α.

Significance levels reflect the seriousness of type-I error in a particular problem, most common values are 0.1, 0.05, 0.01.

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General Procedure for Hypothesis Tests

1. From the problem context, identify the parameter of interest.
2. State the null hypothesis, H_0 .
3. Specify an appropriate alternative hypothesis, H_a.
4. Choose a significance level, α.
5. Determine an appropriate test statistic with the assumption that H_0 is true.
6. State the rejection region for the statistic.
7. Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value.
8. Decide whether or not H_0 should be rejected and report that in the problem context.

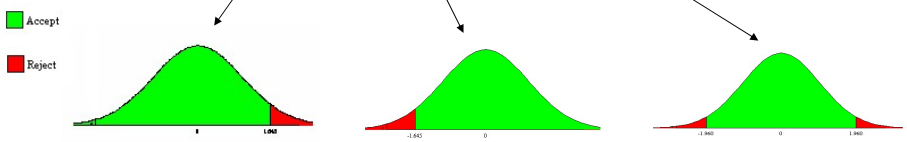
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Tests about a population mean of a normal r.v. with known variance

H_0: $\mu = \mu_0$ Test statistics: $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$

Test statistics is given under the assumption that H_0 is true.

H_1: $\mu > \mu_0$ H_1: $\mu < \mu_0$ H_1: $\mu \neq \mu_0$



R.R. : $z \geq z_\alpha$ R.R. : $z \leq -z_\alpha$ R.R. : $z \leq -z_{\alpha/2}$ or $z \geq z_{\alpha/2}$

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Class exercise: Cigarette manufacturer

A cigarette manufacturer claims that the average nicotine content μ of brand B cigarettes is at most 1.5 mg. It would be unwise to reject the manufacturer's claim without strong contradictory evidence. Suppose the nicotine content is known to be normal with σ=0.2. Assume n=10.

Formulate the hypothesis testing.

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Class exercise: Cigarette manufacturer

A cigarette manufacturer claims that the average nicotine content μ of brand B cigarettes is at most 1.5 mg. **It would be unwise to reject the manufacturer's claim without strong contradictory evidence.** Suppose the nicotine content is known to be normal with σ=0.2. Assume n=10.

Formulate the hypothesis testing.

$H_0: \mu = \mu_0 = 1.5$
$H_a: \mu > \mu_0 = 1.5$

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Class exercise: Cigarette manufacturer

A cigarette manufacturer claims that the average nicotine content μ of brand B cigarettes is at most 1.5 mg. It would be unwise to reject the manufacturer's claim without strong contradictory evidence. Suppose the nicotine content is known to be normal with σ=0.2. Assume n=10.

- Formulate the hypothesis testing.
- Do you reject H_0 if $\bar{x} = 1.596$? Why or why not?

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Class exercise: Cigarette manufacturer

H_0: $\mu = 1.5$
H_a: $\mu > 1.5$

n=10, α=0.05, σ=0.2.

$$\alpha = \text{P\{type I error\}} = \text{P\{reject } H_0, H_0 \text{ is true\}}$$
$$= \text{P}\{\bar{X} > k, \mu = \mu_0\} = \text{P}\{\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} > \frac{k - \mu_0}{\sigma/\sqrt{n}}\}$$

z_α

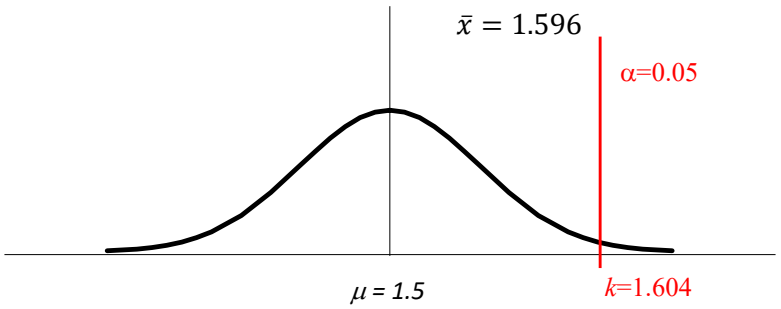
$$k = \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}} = 1.5 + 1.645\frac{0.2}{\sqrt{10}} = 1.604$$

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Example: Cigarette manufacturer

H_0: $\mu = \mu_0 = 1.5$
H_a: $\mu > \mu_0 = 1.5$

$$\alpha = \text{P\{Type I\}} = \text{P\{reject H}_0 \text{ when H}_0 \text{ is true\}} = P\{\bar{X} > k, \mu = 1.5\}$$



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Class exercise: Cigarette manufacturer

A cigarette manufacturer claims that the average nicotine content μ of brand B cigarettes is at most 1.5 mg. It would be unwise to reject the manufacturer's claim without strong contradictory evidence. Suppose the nicotine content is known to be normal with σ=0.2. Assume n=10.

- Formulate the hypothesis testing.
- Do you reject H_0 if $\bar{x} = 1.596$? Why or why not?
- If you cannot reject H_0, what is the probability of a type-II error if μ = 1.6, i.e., $\beta(1.6)$?

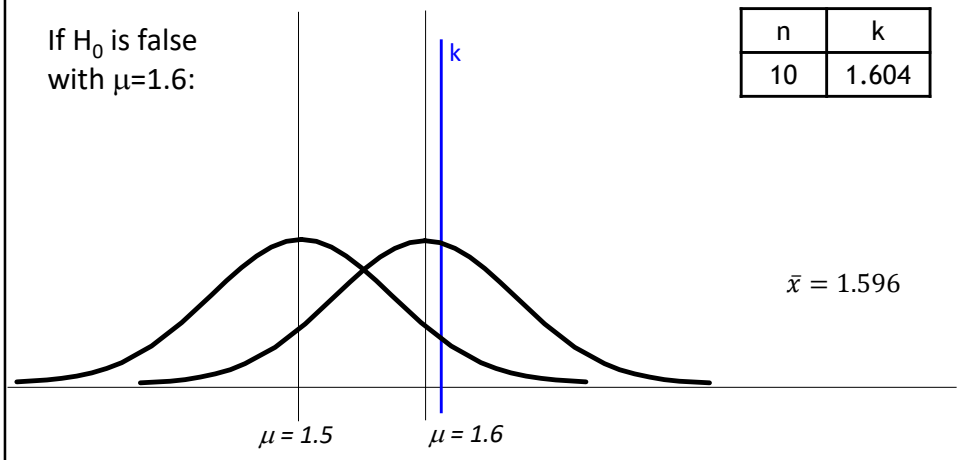
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Example: Cigarette manufacturer

H_0: $\mu = \mu_0 = 1.5$
H_a: $\mu > \mu_0 = 1.5$

If H_0 is false with μ=1.6:



n	k
10	1.604

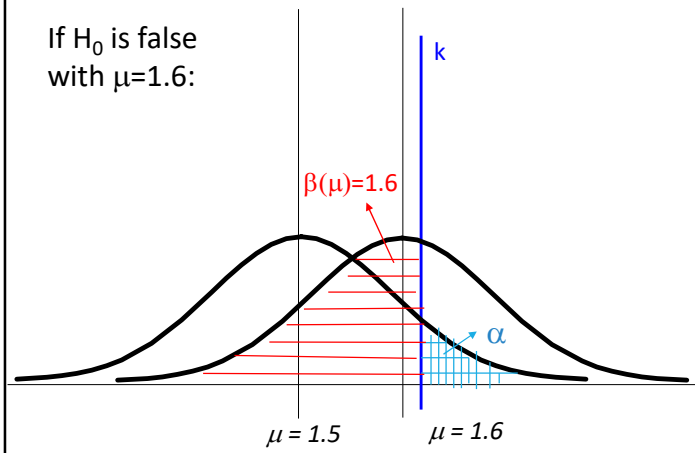
$\bar{x} = 1.596$

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Example: Cigarette manufacturer

H_0: $\mu = \mu_0 = 1.5$
H_a: $\mu > \mu_0 = 1.5$

If H_0 is false with μ=1.6:



n	k
10	1.604

$\bar{x} = 1.596$

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Example: Cigarette manufacturer β and sample size

$$k = \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}}$$

H_0: $\mu = 1.5$
H_a: $\mu > 1.5$

n=10, α=0.05, σ=0.2. What is β(1.6)?

β(1.6) = 0.525

$$\beta(1.6) = \mathrm{P}\{\text{cannot reject } H_0, \mu = 1.6\}$$

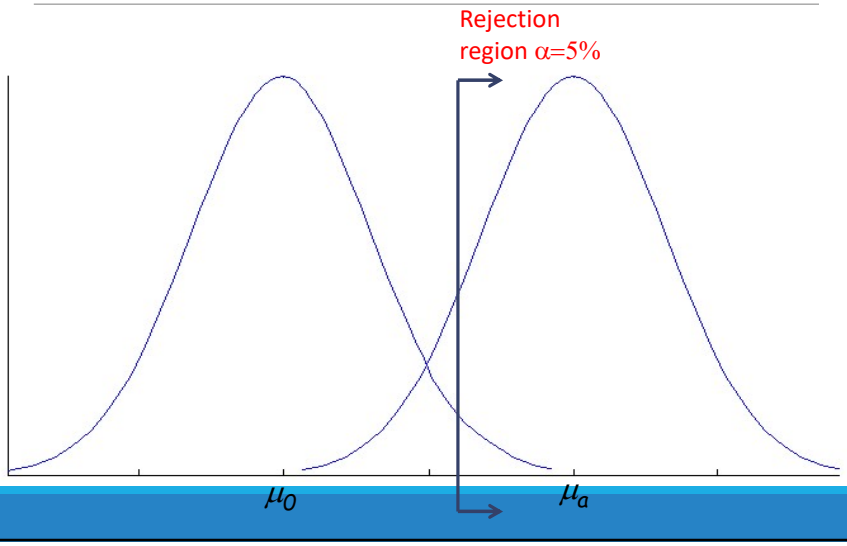
$$= \mathrm{P}\{\bar{X} < k, \mu = \mu_1\} = \mathrm{P}\left\{\frac{\bar{X}-\mu_1}{\sigma/\sqrt{n}} < \frac{\mu_0 + z_\alpha \sigma/\sqrt{n} - \mu_1}{\sigma/\sqrt{n}}\right\}$$

$$= \mathrm{P}\left\{Z < z_\alpha + \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}}\right\}$$

$-z_\beta$

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Illustration of rejection region, α and β



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Proposition

Suppose an experiment and a sample size are fixed and a test statistic is chosen. Then decreasing the size of the rejection region to obtain a smaller value of α results in a larger value of β for any particular parameter value consistent with H_a.

A way to decrease the probability of both errors?

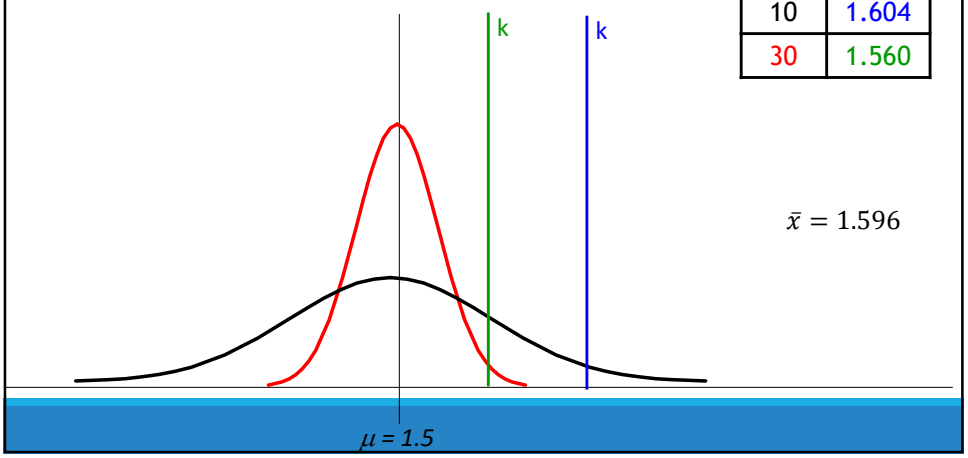
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Effect of sample size: Cigarette manufacturer

H_0: $\mu = \mu_0 = 1.5$
H_a: $\mu > \mu_0 = 1.5$

If H_0 is true and α=0.05, σ=0.2:

n	k
10	1.604
30	1.560



$\bar{x} = 1.596$

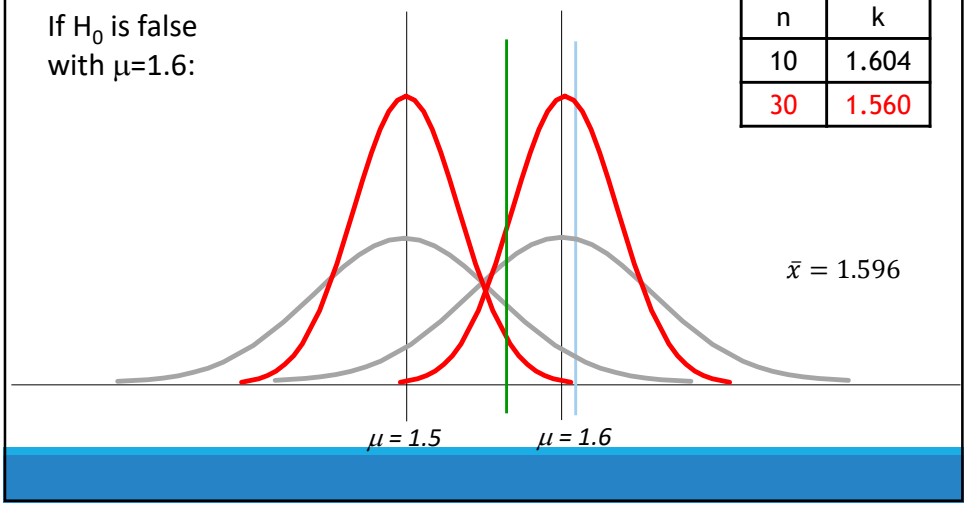
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Effect of sample size: Cigarette manufacturer

H_0: $\mu = \mu_0 = 1.5$
H_a: $\mu > \mu_0 = 1.5$

If H_0 is false with μ=1.6:

n	k
10	1.604
30	1.560



$\bar{x} = 1.596$

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Another approach to hypothesis testing: *P*-values

Hypothesis testing:

$$\alpha \rightarrow \underbrace{\text{Rejection Region}}_{\text{computation of the critical value, k}} \rightarrow \underbrace{\text{conclusion}}_{\text{comparison of } (\bar{x} \text{ and k}) \text{ or } (z \text{ and } z_\alpha)}$$

Significance testing:

$$\bar{x} \rightarrow \underbrace{P\{\text{type I error when } H_0 \text{ is rejected with } k = \bar{x}\}}_{\text{computation of the P-value, P}} \rightarrow \underbrace{\text{conclusion}}_{\text{comparison of P-value and } \alpha}$$

Why? Preference on α may change, *P*-value gives more information.

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P-values

- Definition: The p-value (or observed significance level) is the smallest level of significance at which H_0 would be rejected when a specified test procedure is used on a given data set. Once the p-value has been determined, the conclusion at any particular level α results from comparing the P-value to α:

  1) P-value ≤ α → reject H_0 at level α

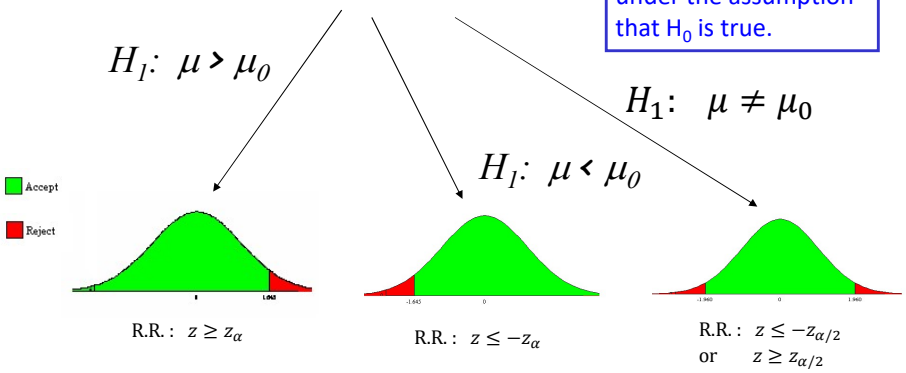
  2) P-value > α → cannot reject H_0 at level α

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Tests about a population mean of a normal r.v. with known variance (Hypothesis Testing)

H_0: $\mu = \mu_0$ Test statistics: $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$

Test statistics is given under the assumption that H_0 is true.

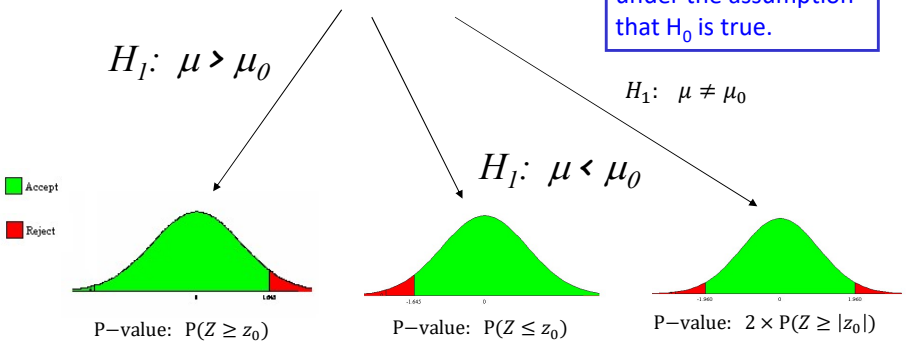


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Tests about a population mean of a normal r.v. with known variance (Significance Testing)

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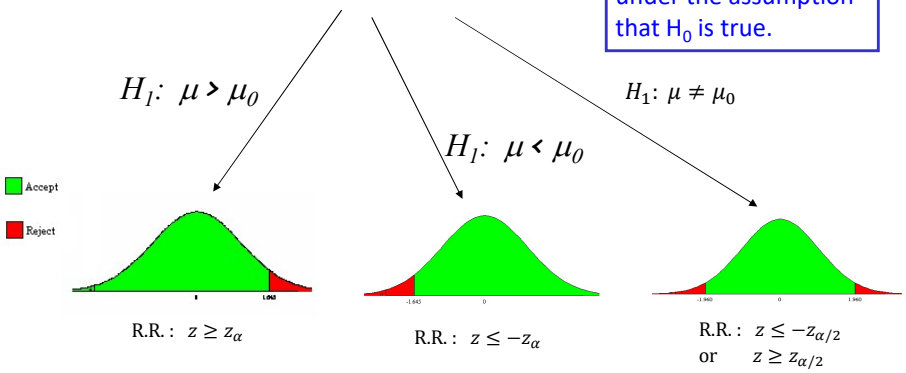


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Tests about a population mean of a r.v. with large sample size

H_0: $\mu = \mu_0$ Test statistics: $Z = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$

Test statistics is given under the assumption that H_0 is true.



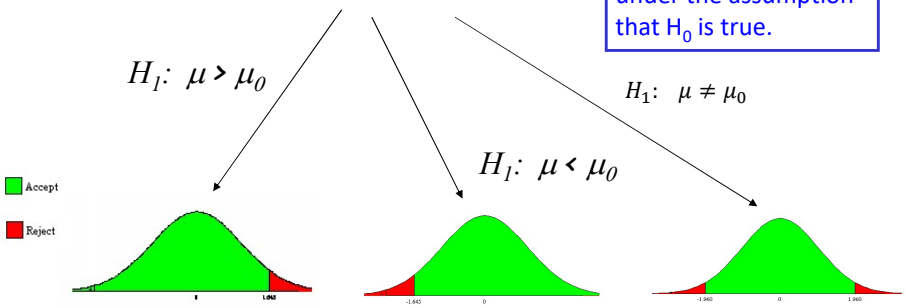
R.R. : $z \geq z_\alpha$ R.R. : $z \leq -z_\alpha$ R.R. : $z \leq -z_{\alpha/2}$ or $z \geq z_{\alpha/2}$

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Tests about a population mean of a r.v. with large sample size (Significance Testing)

H_0: $\mu = \mu_0$ Test statistics: $z_0 = \dfrac{\bar{x} - \mu_0}{S/\sqrt{n}}$

Test statistics is given under the assumption that H_0 is true.



P−value: $P(Z \geq z_0)$ P−value: $P(Z \leq z_0)$ P−value: $2 \times P(Z \geq |z_0|)$

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